Acoustic systems（2nd class）

Yoshikazu Koike

Department of Electronic Engineering

Propagation of the sound in one dimensional system

- Sound wave and magnetic wave
- Velocity potential

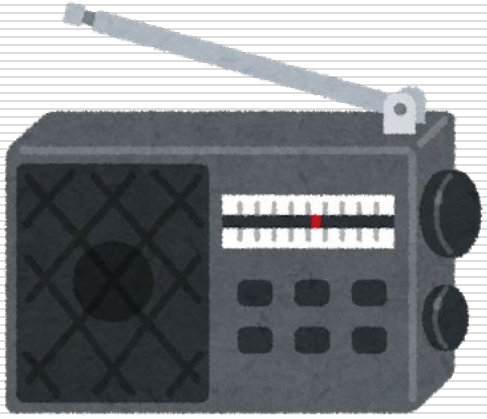
Typical wave phenomena: light and sound

Sound





Radio wave





Comparison of sound waves and electromagnetic waves

	Sound wave	Electromagnetic wave
Propagation	Variation of medium density	Variation of electric field and magnetic field
Kinds of wave	Longitudinal wave Transverse wave	Transverse wave
Propagation medium	Fluid(Gas, water), Solid Not Vacuum	Air, Vacuum Not conductor
Frequency	10Hz to 1GHz	Over 10kHz
Propagation Speed	340 m/s (Air) 1500 m/s (Underwater) 5000 m/s (Metal)	3×10^8 m/s (Vacuum)

Important equation

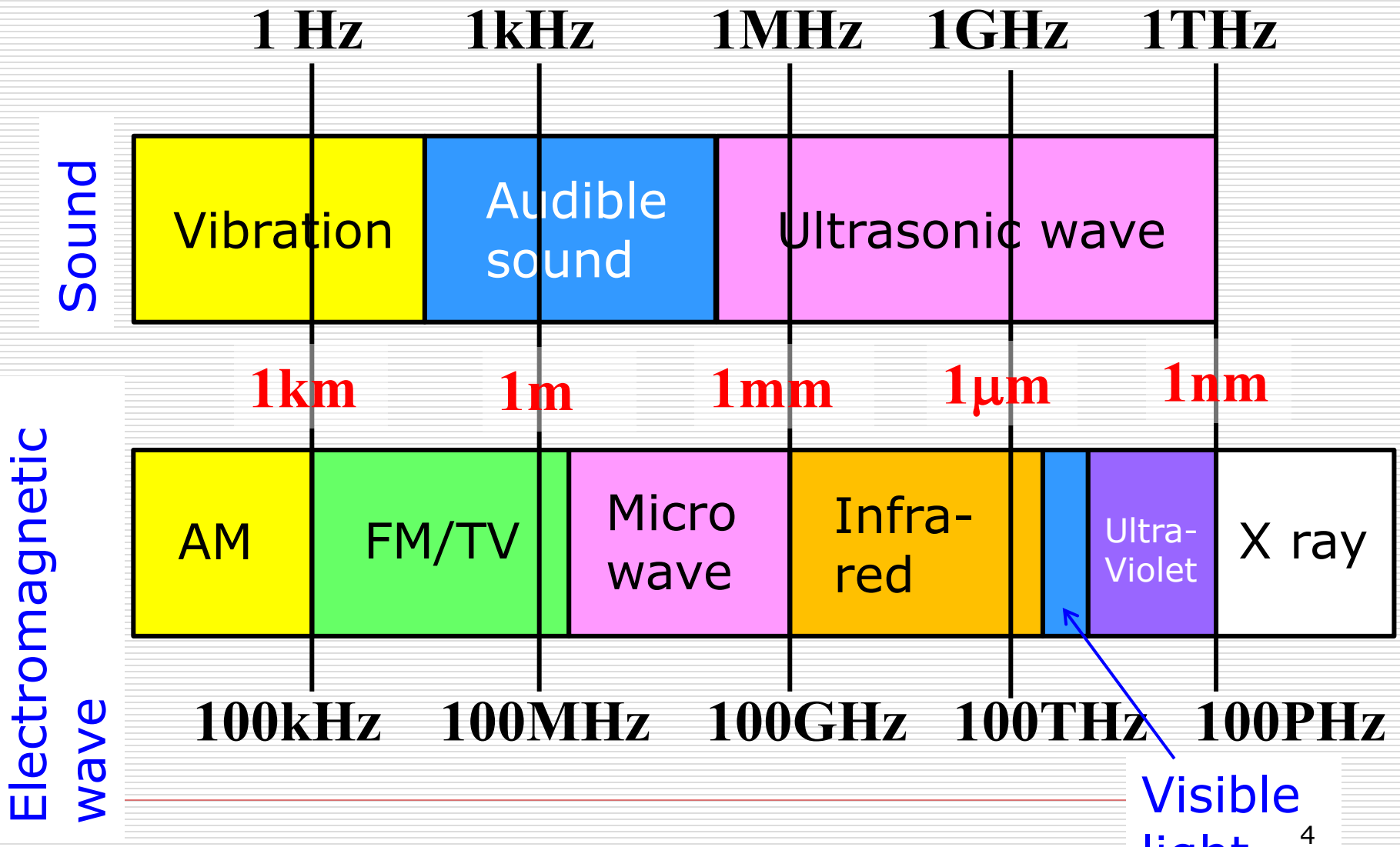
$$c = f\lambda$$

c : propagation speed

f : frequency

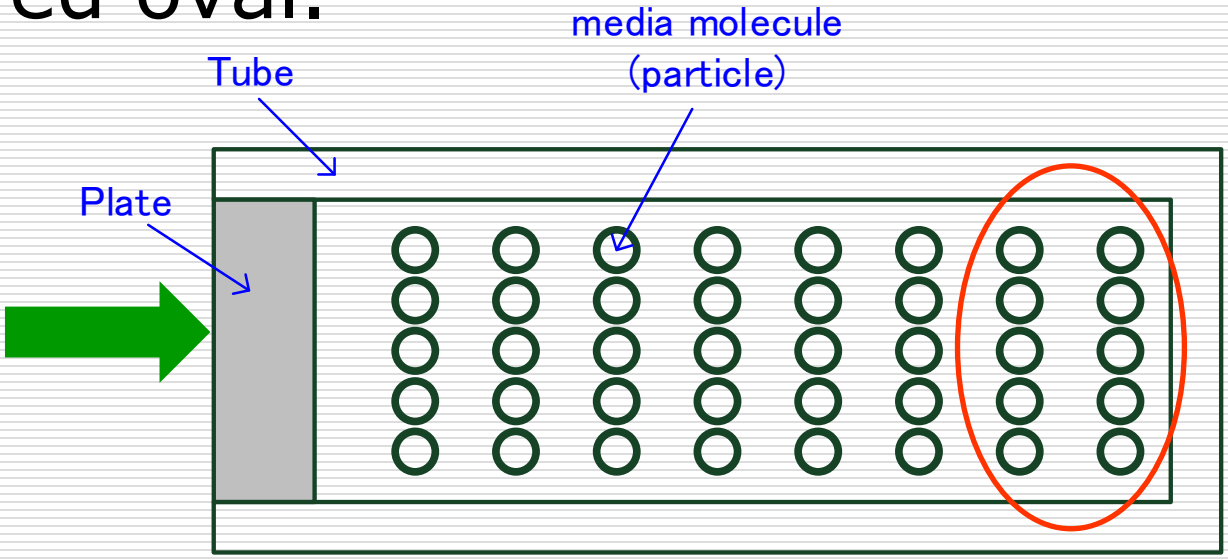
λ : wavelength

Frequency and wavelength of sound waves and electromagnetic waves



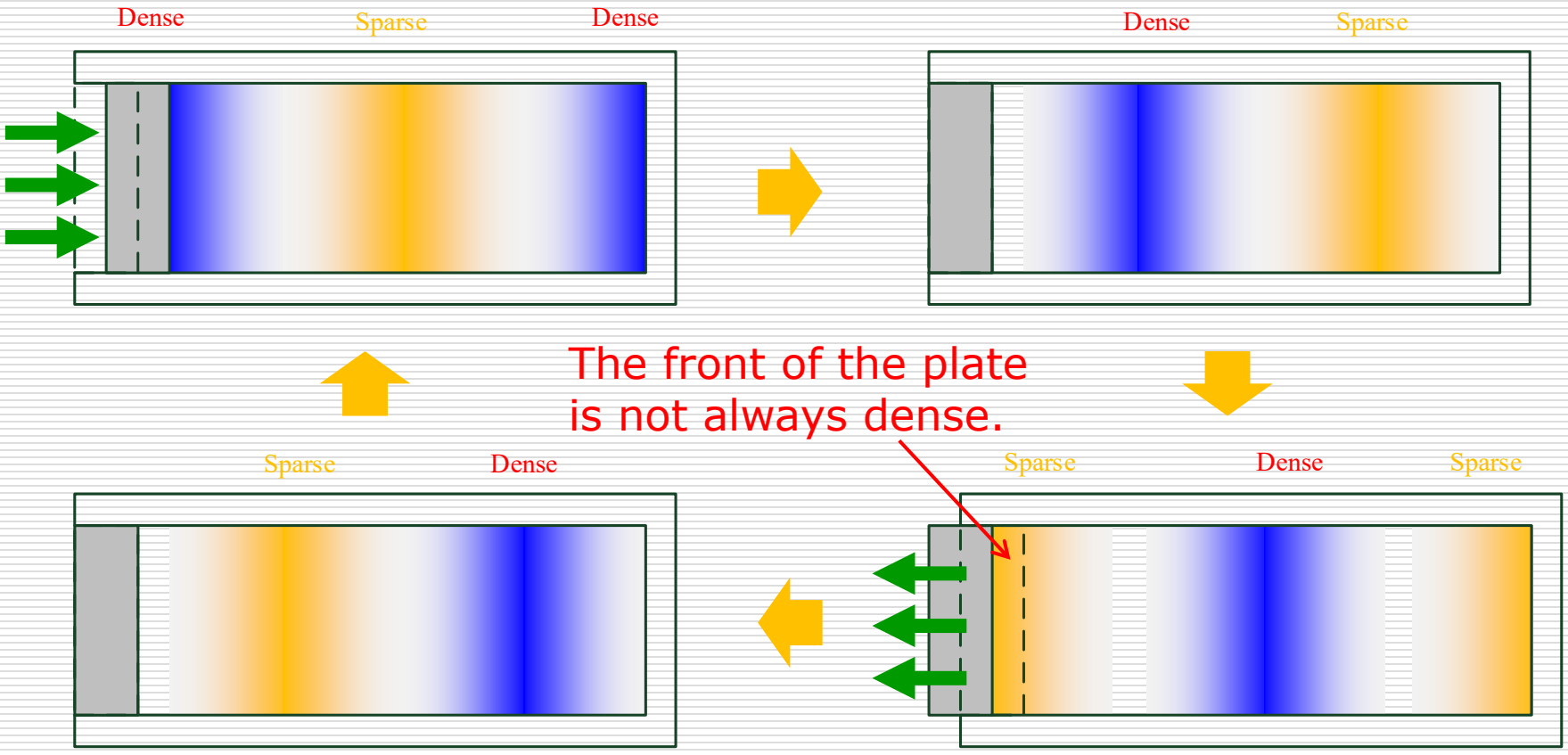
The Generation of Sound Waves

Let's consider the particle motion inside the red oval.



Is the particle motion different when the plate moves quickly compared to when it moves slowly?

Wave propagation

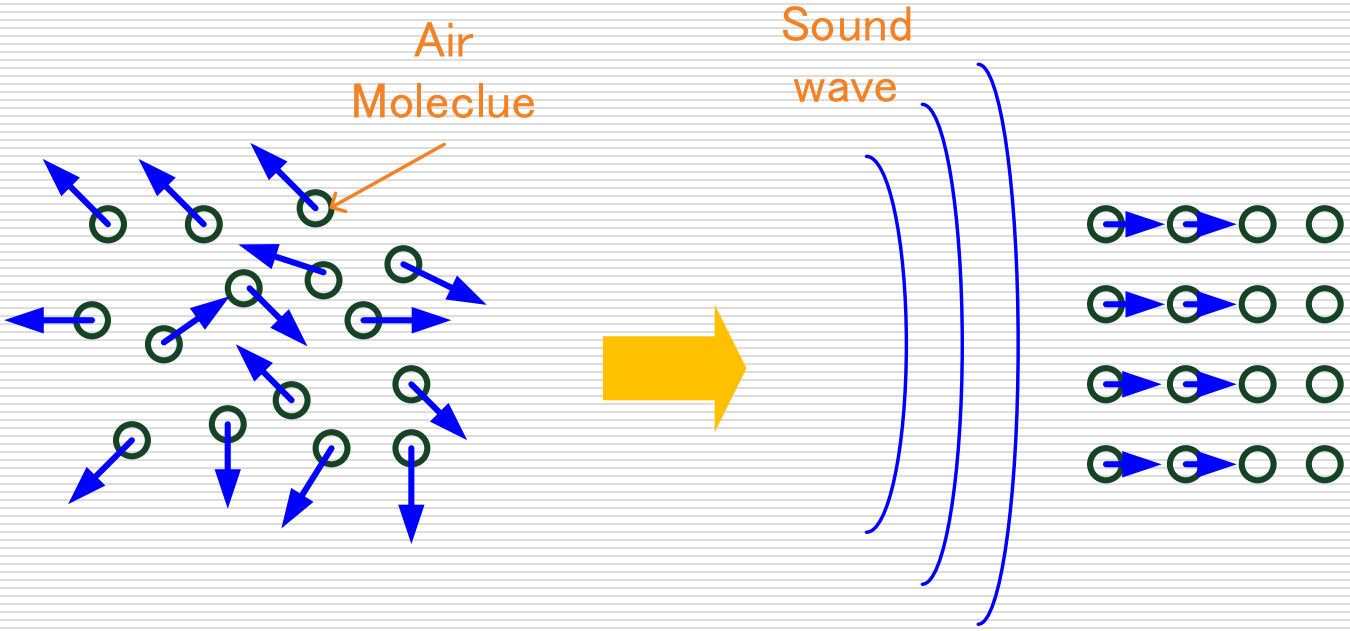


These illustrations are not accurate enough.

Production of sound waves

Air molecules move back and forth in the normal state.

Air molecules move in the direction of sound wave propagation.

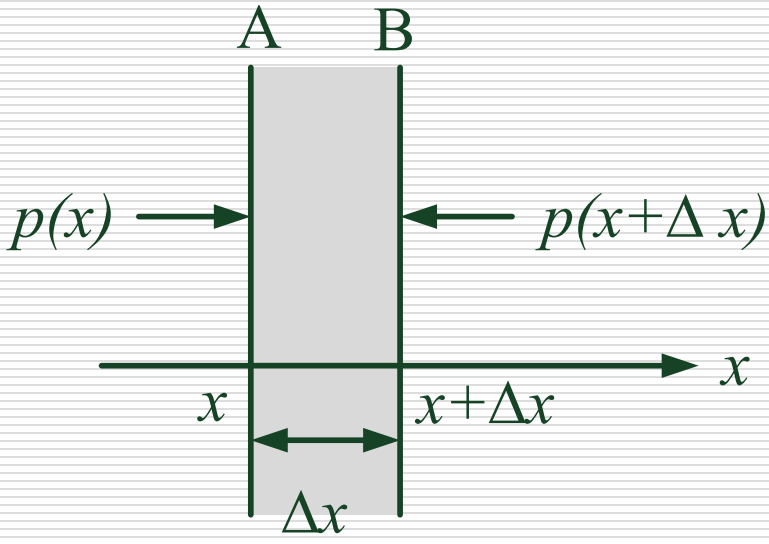


Derivation of the wave equation(1)

motion equation

$$P_0 + p(x+\Delta x,t) = P_0 + p(x,t) + \frac{\partial p}{\partial x}\Delta x$$

$$\rho_0 \frac{\partial v}{\partial t} = -\frac{\partial p}{\partial x}$$



conservation of mass

$$\rho(x+\Delta x)v(x+\Delta x) \cong \rho_0 v + \rho_0 \frac{\partial v}{\partial x}\Delta x + v\frac{\partial \rho}{\partial x}\Delta x$$

$$\frac{\partial \rho}{\partial t} = -\rho_0 \frac{\partial v}{\partial x}$$

Derivation of the wave equation(2)

gas state equation

$$P_0\left(\frac{1}{\rho_0}\right)^{\gamma} = (P_0 + p)\left(\frac{1}{\rho_0 + \rho}\right)^{\gamma}$$

$$p = c^2 \rho \quad , \quad \text{where} \quad c^2 = \gamma \frac{P_0}{\rho_0}$$

Air in sound propagation is caused from adiabatic change.

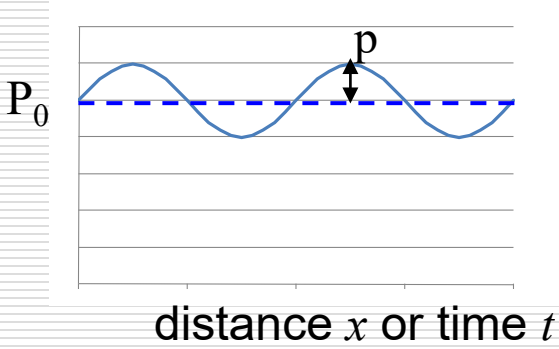
Wave equation in one dimension

Motion equation $\rho_0 \frac{\partial v}{\partial t} = -\frac{\partial p}{\partial x}$

Conservation of mass $\frac{\partial \rho}{\partial t} = -\rho_0 \frac{\partial v}{\partial x}$

Gas state equation $p = c^2 \rho \quad (where \quad c^2 = \frac{\gamma P_0}{\rho_0})$

Static pressure : P_0
Static density : ρ_0, Sound pressure : p,
Deviation of density : ρ,
Particle velocity : v



specific ratio : γ

Wave equation of the sound pressure $\frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = \frac{\partial^2 p}{\partial x^2}$

Wave equation of the particle velocity $\frac{1}{c^2}\frac{\partial^2 v}{\partial t^2} = \frac{\partial^2 v}{\partial x^2}$

Question 1

Try to derive the wave equation,

$$\frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = \frac{\partial^2 p}{\partial x^2}$$

from the following equations.

Motion equation $$\rho_0 \frac{\partial v}{\partial t} = -\frac{\partial p}{\partial x}$$

Conservation of mass $$\frac{\partial \rho}{\partial t} = -\rho_0 \frac{\partial v}{\partial x}$$

Gas state equation $$p = c^2 \rho \quad (where \quad c^2 = \frac{\gamma P_0}{\rho_0})$$

Velocity potential

Sound pressure $\boldsymbol{p}$ and particle velocity $\boldsymbol{v}$ are strongly related. When one - either the sound pressure or the particle velocity - is known, the other is also determined.



Velocity potential $\boldsymbol{\Phi}$ is introduced in order to treat sound pressure and particle velocity simultaneously.

Sound pressure, particle velocity, velocity potintial

Sound pressure $\boldsymbol{p}$ can be obtained by the derivation of velocity potential $\boldsymbol{\Phi}$ in the time variable.

$$p = \rho \frac{\partial \Phi}{\partial t}$$

Density notation:
Instead of ρ_0, a static media density is expressed by ρ.

Particle velocity $\boldsymbol{v}$ can be obtained by the derivation of velocity potential $\boldsymbol{\Phi}$ in the spatial variable.

$$v = -\frac{\partial \Phi}{\partial x}$$

Relation between sound pressure and particle velocity

Wave equation of the sound pressure $\frac{1}{c^2}\frac{\partial^2 p}{\partial t^2}=\frac{\partial^2 p}{\partial x^2}$

Wave equation of the particle velocity $\frac{1}{c^2}\frac{\partial^2 v}{\partial t^2}=\frac{\partial^2 v}{\partial x^2}$

Velocity potential **Φ** is introduced in order to treat sound pressure and particle velocity simultaneously.

Wave equation of the velocity potential $\frac{1}{c^2}\frac{\partial^2 \Phi}{\partial t^2}=\frac{\partial^2 \Phi}{\partial x^2}$

Important relation between ω , k and c

$\boldsymbol{k}$ is wave number. $k = \frac{2\pi}{\lambda}$ $\boldsymbol{c}$ is propagation speed.

ω is angular frequency. $\omega = 2\pi f$

Relation between ω , k and c is the following equation.

$$k = \frac{\omega}{c} \quad \Longleftrightarrow \quad c = f\lambda$$

same relation

Phase from distance or time

Can you calculate what 1[m] in phase of wave phenomena is?

The phase at a certain distance depends on the wave length.

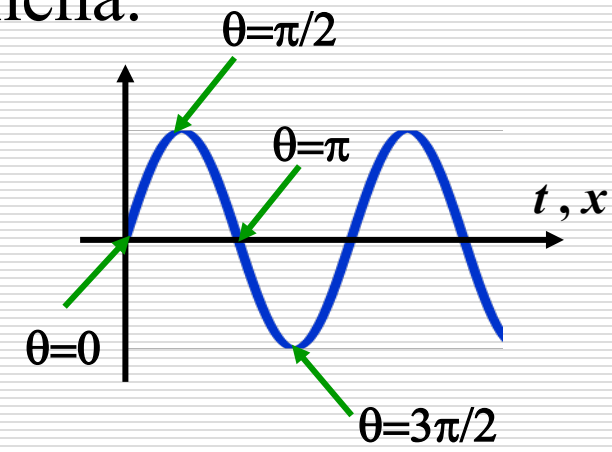
Can you calculate what 1[s] in phase of wave phenomena is?

The phase at a certain time also depends on the period.

Meanings of ω and k.

A phase is important in wave phenomena.

Acoustic phenomena only treats harmonic wave (sinusoidal wave).



k converts the distance x into the phase information θ.

$$\theta = kx$$

ω converts the time t into the phase information θ.

$$\theta = \omega t$$

Helmholtz equation (1)

Time variation in acoustic phenomena is sine function, that is harmonic function.

Therefore, time term is always expressed by $e^{j\omega t}$.

Then, derivative in time variable, that is $\frac{\partial}{\partial t} = j\omega$.

Using velocity potential Φ , sound press is expressed as follows.

$$p = \rho \frac{\partial \Phi}{\partial t} = j\omega\rho\Phi$$

Helmholtz equation (2)

By substituting sound pressure expressed by Φ into wave equation $\frac{1}{c^2}\frac{\partial^2\Phi}{\partial t^2} = \frac{\partial^2\Phi}{\partial x^2}$, the following equation is obtained.

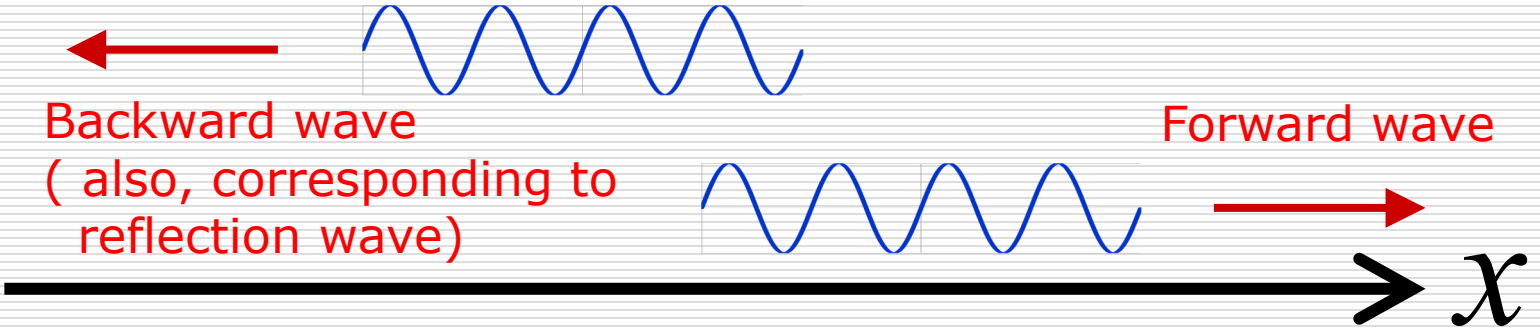
$$-\frac{\omega^2}{c^2}\Phi = \frac{\partial^2\Phi}{\partial x^2} \quad \Rightarrow \quad -k^2\Phi = \frac{\partial^2\Phi}{\partial x^2}$$

Finally, $\frac{\partial^2\Phi}{\partial x^2} + k^2\,\Phi = 0$ ⇐ Helmholtz equation

Solution on wave equation of velocity potential

$$\Phi(x,t) = C_1 e^{j(\omega t - kx)} + C_2 e^{j(\omega t + kx)}$$

Forward wave　　Backward wave



Without reflection, only forward wave propagates.

$$\Phi(x,t) = C_1 e^{j(\omega t - kx)}$$

About time variation

In the acoustic phenomena, only harmonic wave (sinusoidal wave) is treated. Therefore, time variation term is always $C_1 e^{j\omega t}$.

The time variation term is often omitted.

Question 2

The sound wave goes forward only, as written in the equation below. Find the sound pressure p and the particle velocity v, and calculate p/v.

$$\Phi(x,t) = C_1 e^{j(\omega t - kx)}$$

Acoustic characteristic impedance

Acoustic characteristic impedance

Ratio between sound pressure and particle velocity is called as acoustic characteristic impedance (or specific acoustic impedance)

Acoustic characteristic impedance

$$Z_s = \frac{p}{v} = \rho \frac{\omega}{k} = \rho c$$

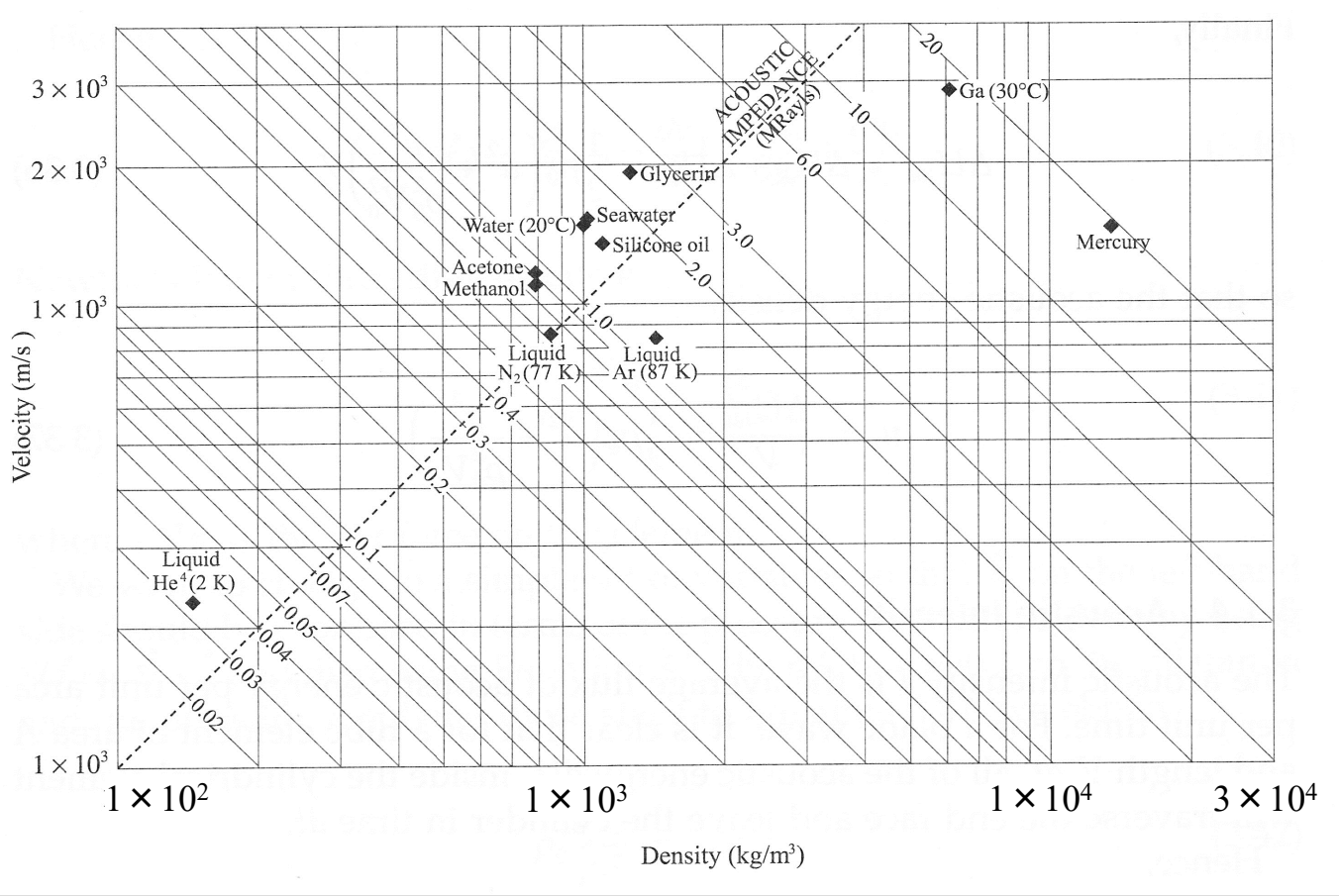
Unit: [Rayls],[Ns/m^3]

Acoustic characteristic impedance is determined by medium.

Typical acoustic characteristic impedance (fluid)

medium	ρ [kg/m³]	c [m/s]	ρc [Pa·s/m]
Water（20℃）	998	1481	1.48×10^{6}
Ethyl alcohol（20℃）	1026	1500	1.54×10^{6}
Air（20℃, 1atm）	1.2	343	4.15×10^{2}
Hydrogen（20℃, 1atm）	0.09	1308	1.18×10^{2}
Helium（0℃, 1atm）	0.179	970	1.73×10^{2}

Relation between density and sound velocity



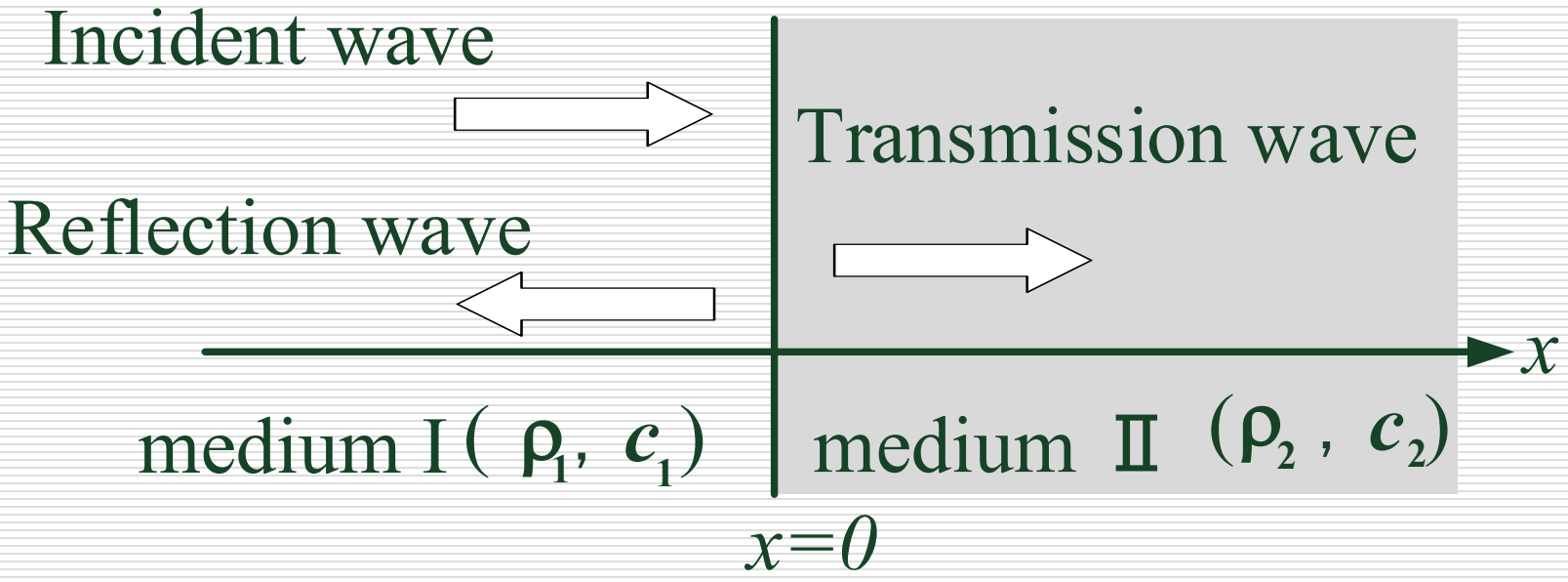
※J.David N.Cheeke: Fundamentals and Applications of Ultrasonic Waves, CRC Press (2002)

Typical acoustic characteristic impedance (solid)

medium	ρ [kg/m³]	c [m/s]	ρc [Pa·s/m]
Steel	7800	5950	27.4×10^6
Gold	19320	3240	62.5×10^6
Glass	2320	5640	13.1×10^6
Quartz	2650	5720	15.16×10^6
Diamond	3520	18000	63.4×10^6
Paper	800	2200	1.76×10^6

Transmission and reflection

Reflection wave and transmission wave occur when propagated wave incidents to the boundary where the acoustic characteristic impedance changes.



Equations at the boundary

Amplitude of incident, reflection and transmission wave indicates as C_1, C_2 and C_3, respectively.

$$\Phi(x,t)=\left(C_1 e^{-jk_1 x}+C_2 e^{jk_1 x}\right)e^{j\omega t} \qquad (x<0)$$

$$\Phi(x,t)=C_3 e^{j(\omega t-k_2 x)} \qquad (x>0)$$

Since sound pressure and particle velocity must be continued at the boundary, the following equations are obtained.

$$\rho_1\left(C_1+C_2\right)=\rho_2 C_3 \quad , \quad k_1\left(C_1-C_2\right)=k_2 C_3$$

Sound pressure reflection coefficient, and sound pressure transmission coefficient

Sound pressure reflection coefficient *R*

$$\frac{\omega\rho_1 C_2}{\omega\rho_1 C_1} = \frac{\rho_2 c_2 - \rho_1 c_1}{\rho_2 c_2 + \rho_1 c_1} \quad (= R)$$

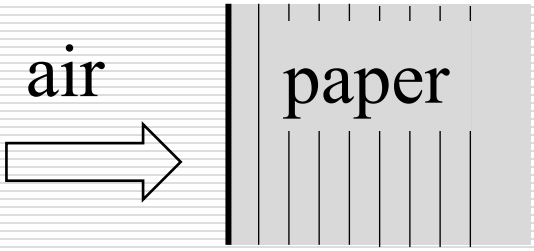
Sound pressure transmission coefficient *T*

$$\frac{\omega\rho_2 C_3}{\omega\rho_1 C_1} = \frac{2\rho_2 c_2}{\rho_2 c_2 + \rho_1 c_1} \quad (= T)$$

Question 3

Calculate the reflection coefficient in the following cases.

1） A sound wave is incident on a stack of paper from the air.



2） A sound wave is incident on air from a stack of paper.

